Foundations of Generative AI (ICT4403)
Assignment 2: Comparative Autoregressive Text Generation

System Architecture, Empirical Evaluation, and Temperature Dynamics

1. Mandatory Submission Identification

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Course & Assignment	Foundations of Generative AI (ICT4403) - Assignment 2
System Architecture Title	Autoregressive Text Generation System using Simple RNN, Bidirectional RNN, LSTM, and GRU
Dataset Name & Source URL	The Adventures of Sherlock Holmes by Sir Arthur Conan Doyle (Project Gutenberg) URL: https://www.gutenberg.org/files/1661/1661-0.txt
GitHub Repository Link	https://github.com/AkCodes23/FGAI_Assignment.git

2. System Overview & Comparable Experimental Setup

To perform a strictly scientific evaluation, all four recurrent architectures were matched in parameter capacity and trained under identical conditions on the causal word-level language modeling task (context sequence length L=25 -> target word w_L+1). The vocabulary comprises 4,378 distinct tokens extracted from Arthur Conan Doyle's The Adventures of Sherlock Holmes. Shared hyperparameters include embedding dimension = 128, hidden dimension = 256, dropout = 0.2, Adam optimizer (lr=0.001), batch size = 64, and gradient norm clipping = 5.0.

3. Quantitative Performance & Operational Efficiency

Model	Parameters	Train Loss	Val Loss	Test Loss	Perplexity	Top-1 Acc (%)	Top-5 Acc (%)	Latency (ms)
Simple RNN	1,915,930	4.2957	5.1250	5.0654	158.44	15.61%	36.90%	2.15
Bidirectional RNN	1,850,394	4.2958	5.0784	5.0280	152.63	16.07%	37.97%	3.20
LSTM	2,607,130	4.6227	4.9441	4.9057	135.06	16.47%	37.95%	1.85
GRU	2,376,730	3.9710	4.9514	4.8810	131.77	18.05%	38.86%	6.94

4. Qualitative Sampling & Temperature Dynamics Analysis

Autoregressive generation was probed across temperatures T in {0.2, 0.7, 1.2} across 3 fixed seed prompts. Distinct-1 (unigram diversity), Distinct-2 (bigram diversity), and 3-Gram Repetition Rate (%) evaluate the transition from greedy exploitation to diverse exploration:

Model	Temperature (T)	Distinct-1	Distinct-2	3-Gram Repetition Rate (%)
Simple RNN	T = 0.2	0.5843	0.8178	22.12%
Simple RNN	T = 0.7	0.7182	0.9531	1.55%
Simple RNN	T = 1.2	0.8454	1.0000	0.00%
Bidirectional RNN	T = 0.2	0.5173	0.7217	27.67%
Bidirectional RNN	T = 0.7	0.8051	0.9713	0.00%
Bidirectional RNN	T = 1.2	0.8510	1.0000	0.00%
LSTM	T = 0.2	0.4895	0.7077	31.27%
LSTM	T = 0.7	0.7568	0.9907	0.00%
LSTM	T = 1.2	0.9138	1.0000	0.00%
GRU	T = 0.2	0.6159	0.8345	12.02%
GRU	T = 0.7	0.7861	0.9829	0.00%
GRU	T = 1.2	0.8762	1.0000	0.00%

5. Qualitative Text Generation Samples (T = 0.7)

[LSTM | T=0.7 | Seed 1]

"it was a dark and stormy night when the next - very man s friend. then i shall imagine that present, do you not not be to the facts. just, and you must do that the cab are. i have heard it."

Distinct-1: 0.7714 | Distinct-2: 1.0000 | Repetition Rate: 0.00%

[LSTM | T=0.7 | Seed 2]

"holmes sat in silence for a few minutes and then said my hand, and i ever given the reading. i understand. he must have been to such the ourselves of his blind. i understand that lord confidence. st. i have made my man of the"

Distinct-1: 0.7692 | Distinct-2: 1.0000 | Repetition Rate: 0.00%

[LSTM | T=0.7 | Seed 3]

"the police had discovered no clue regarding the mystery of the possession. my name was a small minutes of my police there would be deduce with my house, mr. john profession. a own - man was one of the strong cane. i see, mr"

Distinct-1: 0.7297 | Distinct-2: 0.9722 | Repetition Rate: 0.00%

[GRU | T=0.7 | Seed 1]

"it was a dark and stormy night when my father had gone. i am its much, i think, we shall rest. you am sure a drive, but she s very much, said holmes. now, i ve thought that my own"

Distinct-1: 0.8438 | Distinct-2: 1.0000 | Repetition Rate: 0.00%

6. Comparative Architecture Analysis & Findings

- **Superiority of GRU:** The GRU achieved the highest modeling fidelity (Test PPL 131.77, Top-1 Acc 18.05%, Top-5 Acc 38.86%). Its streamlined coupled update/reset gating enables robust gradient transmission over sequence horizons without the parameter overhead of an independent memory cell.
- **LSTM Competitiveness & Speed:** LSTM finished a close second (Test PPL 135.06, Top-1 Acc 16.47%) and exhibited the lowest inference latency (1.85 ms/batch).
- **Simple RNN Limitations:** Simple RNN suffered from severe vanishing gradients, resulting in the highest test perplexity (158.44) and a 22.12% repetition rate at T=0.2 due to mode collapse.
- **Bidirectional RNN Causal Mismatch:** While BiRNN achieved reasonable training loss, it is fundamentally ill-suited for pure autoregressive causal text generation because backward recurrent passes look forward during training, creating a teacher-forcing discrepancy during sequential inference.
- **Temperature Sweet Spot:** T = 0.7 demonstrated the ideal operational balance between lexical diversity (Distinct-2 = 0.98) and syntactic coherence, completely eliminating phrase looping (0.00% 3-gram repetition) for GRU and LSTM.